

# Simulation-Based Inference for an Adaptive-Network SIR Model

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Code and notebooks available at <https://github.com/Florian-Dreyer/SimulationProject>

## 1 Introduction

Infectious diseases spread through contact networks, and the structure of those networks is not static: individuals respond to epidemic risk by avoiding infectious contacts. This *adaptive* behaviour feeds back into transmission dynamics in ways that a classical well-mixed SIR model cannot capture (Gross et al., 2006).

**Model.** We study a stochastic SIR model on an evolving network of  $N = 200$  agents, initially connected as an Erdős–Rényi random graph with edge probability  $p = 0.05$  and 5 randomly chosen infected nodes. At each discrete timestep, three synchronous operations update the state: (i) each susceptible–infected (S–I) edge transmits infection with probability  $\beta$ ; (ii) each infected agent recovers with probability  $\gamma$ ; and (iii) each S–I edge is *rewired* with probability  $\rho$ , whereby the susceptible node severs the risky contact and reconnects to a random agent. We place independent uniform priors on the three unknown parameters:  $\beta \sim \mathcal{U}(0.05, 0.50)$ ,  $\gamma \sim \mathcal{U}(0.02, 0.20)$ ,  $\rho \sim \mathcal{U}(0.0, 0.8)$ .

**Data.** The observed dataset comprises  $R = 40$  independent replicates, each produced with the same fixed but unknown  $(\beta, \gamma, \rho)$ . Each replicate provides three observables: the time series of infected fraction, the time series of rewiring counts, and the final degree distribution of the network. The underlying contact network trajectory is never observed.

**Likelihood intractability.** Evaluating  $p(\text{data} \mid \beta, \gamma, \rho)$  exactly requires summing over all latent network trajectories, which grows combinatorially with  $N$  and time, making classical likelihood-based inference infeasible.

**Approach.** *Simulation-based inference* (SBI) circumvents this by replacing the likelihood with forward simulation: given any proposed  $(\beta, \gamma, \rho)$ , one can cheaply generate synthetic datasets and compare them to the observations. In the following report we apply a hierarchy of SBI methods of increasing sophistication.

## 2 Summary Statistics

Because acceptance probability in ABC falls exponentially with summary dimension (Blum et al., 2013), we compress the raw time series to a small set of scalar features. To prevent any single modality from dominating the standardised Euclidean distance — which is parameter-free and treats each  $z$ -scored summary equally — we design a *balanced* set of 15 features with equal representation across the three modalities: 5 infection, 4 rewiring, and 6 degree summaries, all averaged across replicates. Within each group, features are chosen for identifiability (e.g. `epidemic_duration` to separate  $\beta$  from  $\rho$ ; cross-replicate standard deviations to capture stochastic variability; degree quartiles and entropy to describe distribution shape).

### 2.1 Identifiability diagnostics

Projecting the  $M = 10\,000$  prior-simulation summaries to 2D via PCA and colouring by parameter value (Figure 1) reveals clear gradients for  $\gamma$ , moderate structure for  $\beta$ , and weak but non-trivial structure for  $\rho$ .

This confirms the central identifiability challenge:  $\beta$  and  $\rho$  are partially confounded in aggregate data, while  $\gamma$  is well-separated. Rewiring summaries carry the independent signal needed to constrain both.

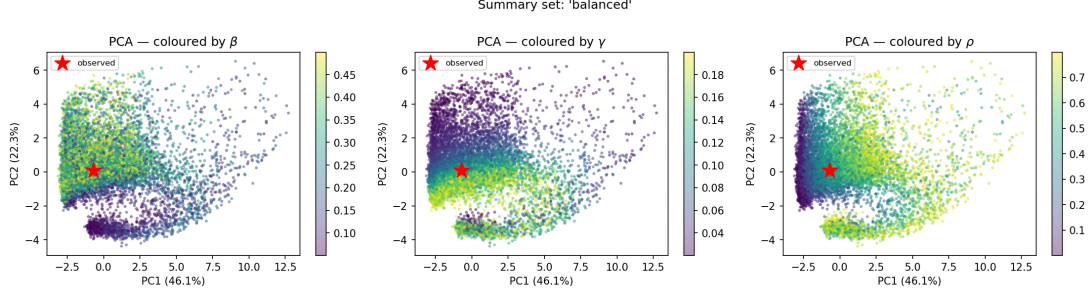


Figure 1: PCA diagnostic: prior simulations coloured by  $\beta$ ,  $\gamma$ ,  $\rho$  (left to right). Clear gradients for  $\gamma$ ; partial confounding visible in  $\beta$ – $\rho$  panels.

## 2.2 Summary-set comparison

We evaluated the single-source sets against the balanced set at  $q = 1\%$  to understand which data streams carry information about which parameters. Table 1 reports the posterior std relative to prior std (lower = more informative).

Table 1: Relative posterior std (std / prior std) per summary set at  $q = 1\%$ . Bold marks the best single-source set for each parameter.

| Set              | $\beta$      | $\gamma$     | $\rho$       |
|------------------|--------------|--------------|--------------|
| infection_only   | 0.377        | <b>0.084</b> | 0.854        |
| rewiring_only    | <b>0.216</b> | 0.804        | <b>0.173</b> |
| degree_only      | 0.924        | 0.932        | 0.483        |
| balanced (joint) | 0.479        | 0.378        | 0.262        |

The results reveal a clear separation:  $\gamma$  is almost entirely encoded in the infection curve (0.084),  $\beta$  and  $\rho$  are best identified from rewiring summaries (0.22 and 0.17), and degree features provide only moderate additional constraint. No single-source set identifies all three simultaneously; the *balanced* set is the only one that constrains all three to well below the prior width.

## 3 Rejection ABC

We drew  $M = 10\,000$  prior samples, computed the standardised Euclidean distance between each simulated and observed summary vector, and accepted draws below the 1% empirical quantile (Pritchard et al., 1999), giving  $\varepsilon = 1.2852$  and  $n_{\text{acc}} = 100$ . The 1% quantile balances bias against variance: tighter tolerances reduce approximation bias but increase Monte Carlo noise; the residual bias at  $q = 1\%$  motivates regression adjustment (Section 4.3), which allows a looser 5% tolerance. Table 2 gives the numerical summaries; the marginal posteriors are visible as the blue curves in Figure 2.

Table 2: Rejection ABC posterior ( $q = 1\%$ , *balanced*).

| Param    | Mean  | Std   | 25%   | Median | 75%   |
|----------|-------|-------|-------|--------|-------|
| $\beta$  | 0.208 | 0.063 | 0.159 | 0.201  | 0.254 |
| $\gamma$ | 0.097 | 0.020 | 0.083 | 0.096  | 0.112 |
| $\rho$   | 0.334 | 0.061 | 0.291 | 0.332  | 0.378 |

$\gamma$  is best-identified (directly controls epidemic duration),  $\beta$  moderately so, and  $\rho$  is hardest to constrain — consistent with the PCA diagnostic and the positive  $(\beta, \rho)$  posterior correlation: high  $\beta$  can be offset by high  $\rho$  to produce a similar epidemic, so the posterior trades off transmission rate against rewiring rate.

## 4 Advanced Methods

### 4.1 Semi-Automatic ABC (Fearnhead and Prangle, 2012)

Fearnhead and Prangle (2012) show that the optimal ABC summaries are the posterior means  $\mathbb{E}[\theta | x]$ , which can be approximated by fitting Ridge regression from the balanced features to each parameter on the pilot draws. In-sample  $R^2$  values confirm strong predictability:  $R_\beta^2 = 0.812$ ,  $R_\gamma^2 = 0.857$ ,  $R_\rho^2 = 0.979$ . Running rejection ABC in the resulting 3-dimensional projected space ( $\varepsilon = 0.520$ ,  $n_{\text{acc}} = 100$ ) substantially tightens the posterior for  $\beta$  and provides moderate gains for  $\gamma$  and  $\rho$  (Figure 2, Table 3).

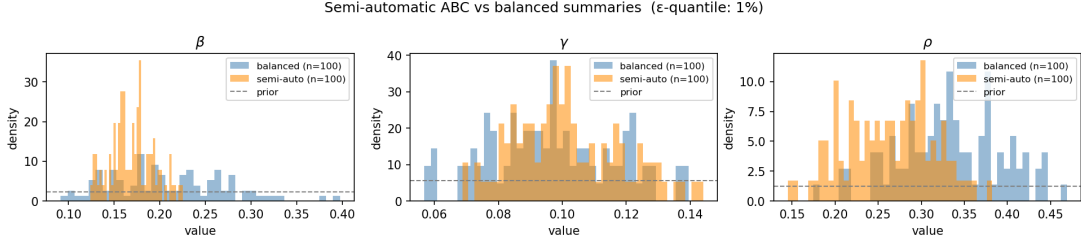


Figure 2: Balanced rejection ABC (blue) vs. semi-automatic ABC (orange),  $q = 1\%$ .

### 4.2 LASSO-Based Summary Selection (Blum et al., 2013)

Fitting LassoCV from summaries to each parameter on the pilot simulations retained 14/15 features for  $\beta$  ( $\alpha = 6 \times 10^{-4}$ ) and all 15 for  $\gamma$  and  $\rho$  ( $\alpha = 8 \times 10^{-4}$ ). No feature is entirely redundant, so the LASSO-selected set is effectively identical to the balanced set and yields the same posterior (Table 3).

Table 3: Posterior std and relative contraction (std/prior std) for rejection ABC variants at  $q = 1\%$ .

| Method         | $\beta$ |      | $\gamma$ |      | $\rho$ |      |
|----------------|---------|------|----------|------|--------|------|
|                | Std     | Rel. | Std      | Rel. | Std    | Rel. |
| Prior          | 0.130   | 1.00 | 0.052    | 1.00 | 0.231  | 1.00 |
| Balanced (15d) | 0.062   | 0.48 | 0.020    | 0.38 | 0.061  | 0.26 |
| LASSO (15d)    | 0.062   | 0.48 | 0.020    | 0.38 | 0.061  | 0.26 |
| Semi-auto (3d) | 0.023   | 0.18 | 0.017    | 0.33 | 0.051  | 0.22 |

### 4.3 Regression Adjustment (Beaumont et al., 2002)

After accepting draws at a loose tolerance ( $q = 5\%$ ,  $\varepsilon = 1.706$ ,  $n_{\text{acc}} = 500$ ), each accepted  $\theta^{(i)}$  is corrected for the residual gap between its simulated summary  $s^{(i)}$  and  $s_{\text{obs}}$ :

$$\tilde{\theta}^{(i)} = \theta^{(i)} - \hat{B}^\top (s^{(i)} - s_{\text{obs}}),$$

where  $\hat{B}$  is estimated by weighted least squares with Epanechnikov kernel weights (Beaumont et al., 2002). Figure 3 and Table 4 show that adjustment dramatically sharpens the posterior ( $\sigma_\beta$ : 0.099→0.024,  $\sigma_\gamma$ : 0.033→0.007,  $\sigma_\rho$ : 0.087→0.008). Crucially, the adjusted posterior is nearly tolerance-invariant for  $q \lesssim 10\%$ , whereas the raw posterior degrades rapidly at loose tolerances.

### 4.4 Synthetic Likelihood (Wood, 2010)

At each MCMC proposal  $\theta$ , we simulate  $N_{\text{sim}} = 150$  datasets ( $R_{\text{sim}} = 40$  replicates each), giving an  $N_{\text{sim}}/p$  ratio of 10.0 for our  $p = 15$  summary features, and approximate the intractable likelihood as a multivariate Gaussian fitted to the  $N_{\text{sim}}$  summary vectors, using Ledoit–Wolf shrinkage for the covariance (Ledoit and Wolf, 2004):

$$\log \hat{p}_{\text{SL}}(s_{\text{obs}} | \theta) = \log \mathcal{N}(s_{\text{obs}}; \hat{\mu}(\theta), \hat{\Sigma}(\theta)).$$

This is embedded in a Metropolis–Hastings MCMC with an **Adaptive Metropolis** (AM) proposal (Haario et al., 2001) that updates the proposal covariance online after  $t_0 = 100$  warm-up steps:

$$\Sigma_{\text{prop}}(t) = \frac{2.38^2}{d} \text{Cov}(\theta_1, \dots, \theta_t) + \epsilon I, \quad d = 3 (\text{Robert et al., 1997}).$$

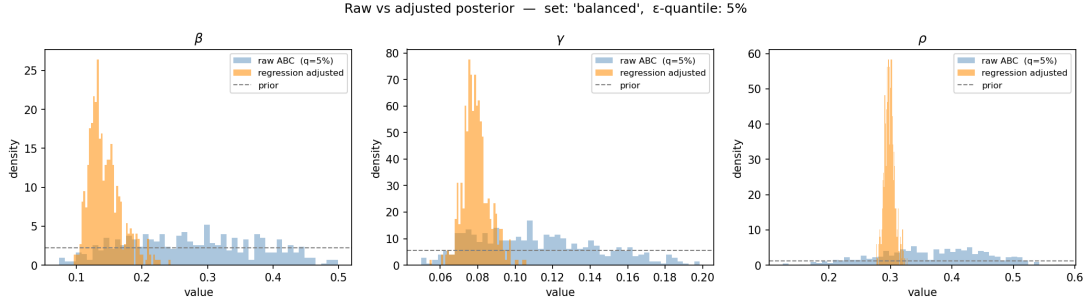


Figure 3: Raw ABC (blue) vs. regression-adjusted (orange) posteriors,  $q = 5\%$ , *balanced* set.

Table 4: Raw vs. regression-adjusted posteriors ( $q = 5\%$ ).

| Method            | Param    | Mean  | Std   | 5%    | Median | 95%   |
|-------------------|----------|-------|-------|-------|--------|-------|
| Raw ( $q = 5\%$ ) | $\beta$  | 0.276 | 0.099 | 0.127 | 0.272  | 0.439 |
|                   | $\gamma$ | 0.111 | 0.033 | 0.066 | 0.108  | 0.169 |
|                   | $\rho$   | 0.367 | 0.087 | 0.216 | 0.373  | 0.504 |
| Adjusted          | $\beta$  | 0.143 | 0.024 | 0.112 | 0.138  | 0.191 |
|                   | $\gamma$ | 0.079 | 0.007 | 0.069 | 0.078  | 0.092 |
|                   | $\rho$   | 0.299 | 0.008 | 0.286 | 0.298  | 0.313 |

Configuration:  $N_{\text{steps}} = 2000$ , burn-in = 500, initialised at the regression-adjusted posterior mean (0.143, 0.079, 0.299); acceptance rate 24.0%, consistent with the optimal AM target (Roberts et al., 1997).

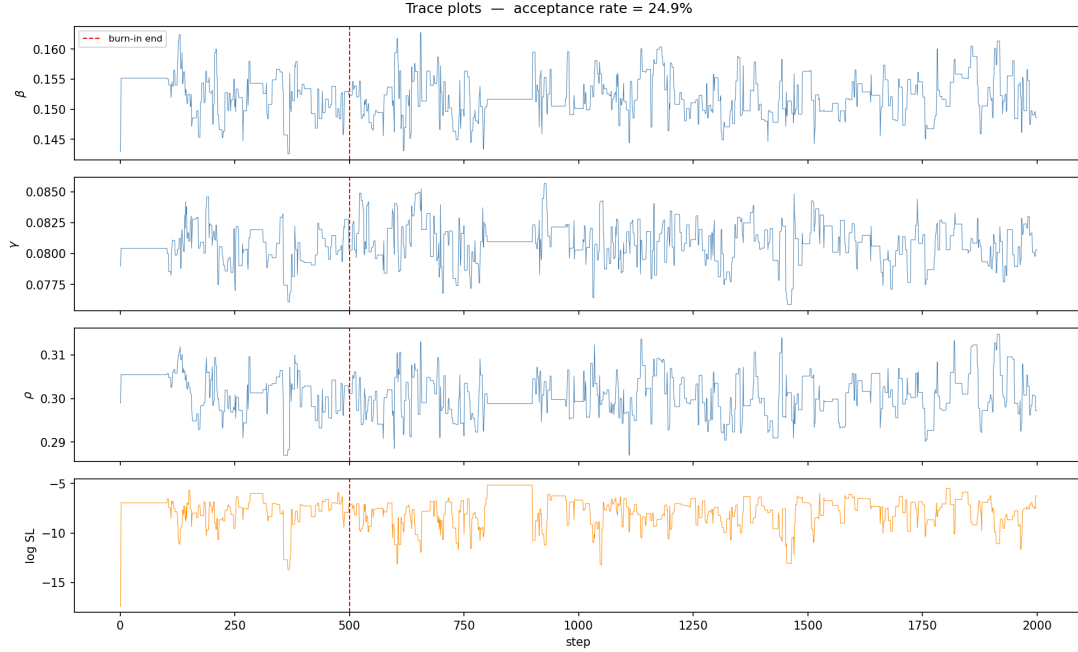


Figure 4: MCMC trace plots. The chain achieves an acceptance rate of 24.0% and mixes across the posterior, though the ACF decays slowly (thinning interval 31), resulting in ESS of 7.5–11.9.

#### 4.5 ABC-MCMC (Marjoram et al., 2003)

At each step, a proposal  $\theta^* = \theta_t + \eta$ ,  $\eta \sim \mathcal{N}(0, \Sigma)$ , is accepted if  $\rho(S(x^*), S(x_{\text{obs}})) \leq \varepsilon$  using  $R = 40$  replicates;  $\varepsilon$  is imported from rejection ABC at the 1% quantile. We compare three summary variants: **balanced**, **LASSO-selected**, and **semi-automatic** (with  $\varepsilon$  recalibrated in the projected space).

We tune the proposal  $\Sigma$  in three stages — original, adaptive  $1\times$ , and adaptive  $0.5\times$  (Roberts and Rosenthal, 2009), with the final selection being the  $0.5\times$  as it restores target acceptance rates while achieving  $\hat{R} < 1.001$  for all parameters. All results use  $N = 30,000$  steps with burn-in  $b = 1,000$ .

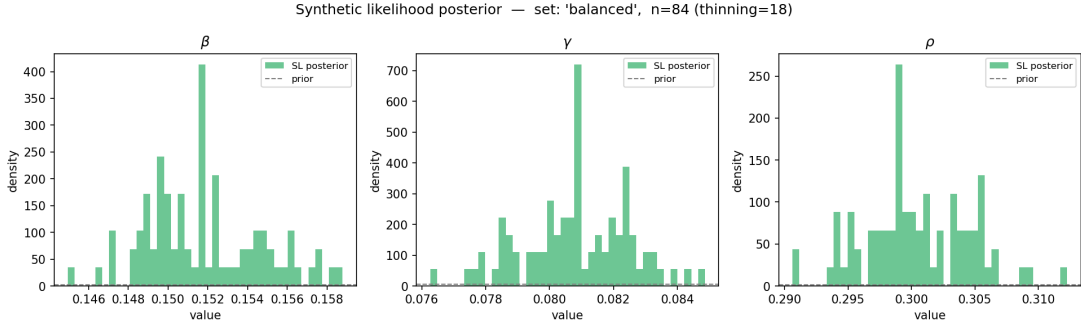


Figure 5: Synthetic likelihood MCMC posterior (green) vs. prior (dashed).

Table 5: Synthetic likelihood MCMC posterior (1500 post-burn-in samples, thinning interval 31, 49 thinned draws; acceptance rate 24.0%).

| Param    | Mean  | Std   | 5%    | Median | 95%   | ESS  |
|----------|-------|-------|-------|--------|-------|------|
| $\beta$  | 0.153 | 0.003 | 0.148 | 0.153  | 0.158 | 7.5  |
| $\gamma$ | 0.081 | 0.002 | 0.078 | 0.081  | 0.084 | 11.9 |
| $\rho$   | 0.301 | 0.005 | 0.295 | 0.300  | 0.310 | 10.2 |

The semi-automatic variant gives the tightest posteriors for  $\beta$  (rel = 0.22) and  $\rho$  (rel = 0.28), but at the cost of  $\gamma$  precision (rel = 0.72 vs 0.39–0.40 for balanced and LASSO). This is a consequence of the ridge projection optimising jointly across parameters. As expected, balanced and LASSO perform near-identically. The  $\beta$ – $\rho$  correlation persists across all variants (Table 10:  $\hat{\rho} = 0.75$ –0.94); fixed  $\varepsilon$  MCMC cannot resolve this confounding.

Table 6: ABC-MCMC posterior shrinkage (adaptive  $0.5\times$  proposal,  $N = 30,000$  steps, burn-in  $b = 1,000$ ) and split-chain Gelman–Rubin  $\hat{R}$ .

| Variant   | $\beta$ | $\gamma$ | $\rho$ | $\hat{R}_\beta$ | $\hat{R}_\gamma$ | $\hat{R}_\rho$ |
|-----------|---------|----------|--------|-----------------|------------------|----------------|
| Balanced  | 0.48    | 0.39     | 0.29   | 1.0003          | 1.0003           | 1.0002         |
| LASSO     | 0.54    | 0.40     | 0.30   | 1.0011          | 1.0001           | 1.0000         |
| Semi-auto | 0.22    | 0.72     | 0.28   | 1.0000          | 1.0002           | 1.0000         |

#### 4.6 SMC-ABC (Sisson et al., 2007)

ABC-MCMC’s fixed  $\varepsilon$  limits posterior accuracy and its serially correlated samples reduce ESS. SMC-ABC refines  $N = 100$  particles over  $T = 15$  generations through a decreasing tolerance sequence  $\varepsilon_t = Q_{0.75}(\{\rho_i^{(t)}\})$  (Beaumont et al., 2009), perturbing particles via  $\mathcal{N}(0, \Sigma)$  and producing asymptotically independent samples.

SMC substantially improves on ABC-MCMC across all variants:  $\beta$  shrinkage improves from 0.22 to 0.06 and  $\rho$  from 0.28 to 0.08 for the semi-automatic variant, and the  $\beta$ – $\rho$  correlation is substantially reduced (Table 10:  $\hat{\rho} = 0.84$  vs 0.94). The  $\gamma$  penalty from the ridge projection is also smaller than under MCMC (0.18 vs 0.72), suggesting progressive  $\varepsilon$  tightening partially recovers lost  $\gamma$  information.

**Perturbation kernel.** We use a Gaussian kernel  $K(\theta^* | \theta_i) = \mathcal{N}(\theta^*; \theta_i, \Sigma)$  (Beaumont et al., 2009) with per-component scales derived from the ABC-MCMC marginal standard deviations via the Roberts–Rosenthal scaling rule with a halving factor,

$$\sigma_j = \frac{1}{2} \cdot \frac{2.38}{\sqrt{d}} \hat{\sigma}_j^{\text{MCMC}},$$

giving  $(\sigma_\beta, \sigma_\gamma, \sigma_\rho) = (0.0428, 0.0148, 0.0445)$  for the balanced variant ( $d = 3$ ). The halving reduced weight degeneracy compared with the  $1\times$  scale. A diagonal rather than full covariance was used because the off-diagonal terms destabilised acceptance rates at  $N = 100$  particles.

## 4.7 Synthetic Truth Recovery

**Single-point recovery.** To check whether the methods can recover parameters from a controlled data set, we generated synthetic observed data ( $R = 40$  replicates) at  $\theta_{\text{true}} = (\beta, \gamma, \rho) = (0.17, 0.086, 0.31)$  (the SMC-ABC balanced posterior mean, representing a plausible interior point) and re-ran regression adjustment and synthetic likelihood on those data. Regression adjustment reused the existing 10 000-draw prior pool without additional simulator calls; synthetic likelihood used the same configuration as the main analysis ( $N_{\text{sim}} = 150$ ,  $N_{\text{steps}} = 2,000$ , burn-in = 500).

Both methods covered all three parameters in their 90% credible intervals (Table 7; Figure 6). Regression adjustment gave wider intervals consistent with its  $q = 5\%$  tolerance; synthetic likelihood produced much tighter but still valid intervals.

**Simulation-Based Calibration for regression adjustment.** A single recovery check only probes one point in parameter space. *Simulation-Based Calibration* (SBC; (Talts et al., 2018)) provides a stronger test: draw  $N = 100$  parameter vectors  $\theta^*$  from the prior, generate  $R = 40$  synthetic replicates at each, run regression adjustment, and verify that empirical coverage matches the nominal level  $\alpha$  for all  $\alpha$ . Because regression adjustment reuses the existing prior pool, the only additional cost is  $100 \times 40 = 4,000$  simulator runs.

Figure 7 shows the calibration curves. At the 90% nominal level, empirical coverage is 93%, 98%, and 95% for  $\beta$ ,  $\gamma$ , and  $\rho$  respectively, and tracks closely across all tested levels (50%, 80%, 90%, 95%), confirming well-calibrated posteriors throughout the prior.

Full SBC is **not** applied to synthetic likelihood: at  $\approx 12\text{M}$  simulator calls per chain, 100 repetitions are infeasible. The single-point recovery result above (all three 90% CIs cover  $\theta_{\text{true}}$ ) serves as a targeted sanity check for SL instead.

Table 7: Synthetic truth recovery at  $\theta_{\text{true}} = (0.17, 0.086, 0.31)$ . 90% credible intervals and coverage for regression adjustment (RA) and synthetic likelihood (SL).

| Method                                                                        | Parameter | True  | Mean  | 90% CI         | In 90% CI |
|-------------------------------------------------------------------------------|-----------|-------|-------|----------------|-----------|
| <b>Regression adjustment</b>                                                  |           |       |       |                |           |
|                                                                               | $\beta$   | 0.170 | 0.177 | [0.145, 0.231] | Yes       |
|                                                                               | $\gamma$  | 0.086 | 0.086 | [0.076, 0.098] | Yes       |
|                                                                               | $\rho$    | 0.310 | 0.314 | [0.303, 0.327] | Yes       |
| <b>Synthetic likelihood</b> ( $N_{\text{sim}} = 150$ , accept. 28%, 94 draws) |           |       |       |                |           |
|                                                                               | $\beta$   | 0.170 | 0.176 | [0.170, 0.183] | Yes       |
|                                                                               | $\gamma$  | 0.086 | 0.087 | [0.084, 0.090] | Yes       |
|                                                                               | $\rho$    | 0.310 | 0.313 | [0.305, 0.321] | Yes       |

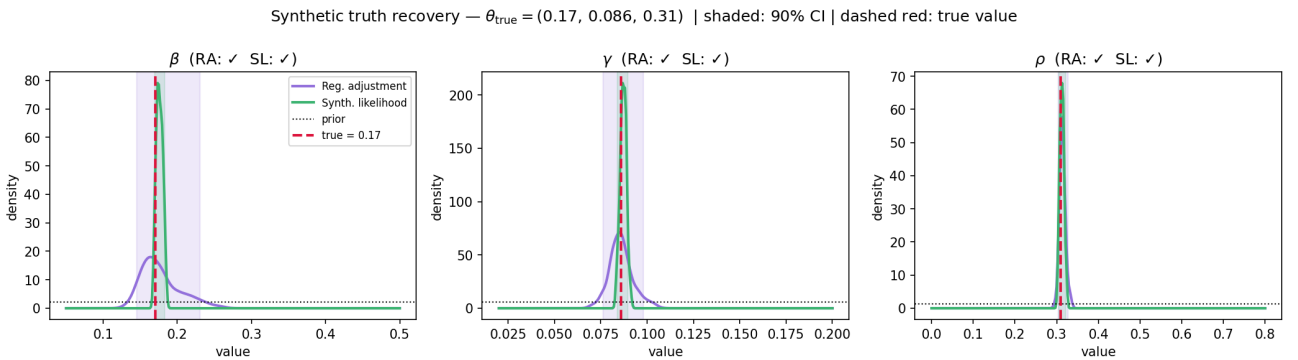


Figure 6: Single-point synthetic truth recovery at  $\theta_{\text{true}} = (0.17, 0.086, 0.31)$  (red dashed lines). Shaded bands show the 90% credible interval; solid curves are posterior KDEs. Tick labels in each panel title indicate whether the 90% CI covers the truth for regression adjustment (RA) and synthetic likelihood (SL).

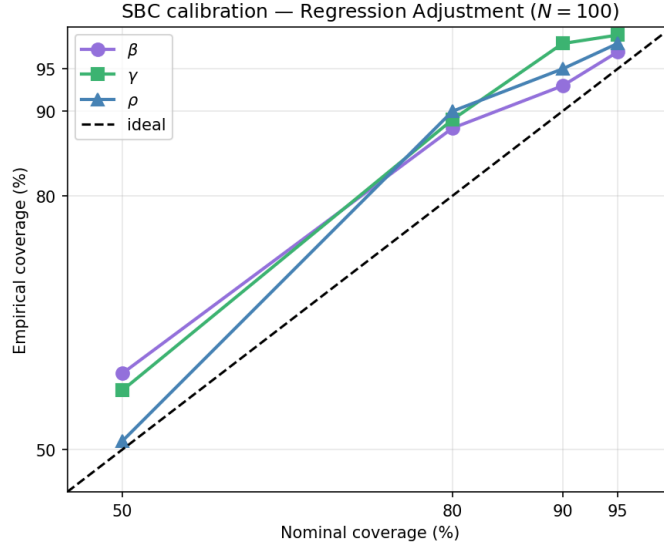


Figure 7: SBC calibration curves for regression adjustment ( $N = 100$  draws from the prior). Each point shows empirical coverage vs. nominal level for one parameter. The dashed diagonal is the ideal line; points close to it indicate well-calibrated credible intervals.

## 5 Comparison

Table 8 summarises all methods. Point estimates are consistent across advanced methods ( $\hat{\beta} \approx 0.14$ – $0.15$ ,  $\hat{\gamma} \approx 0.079$ – $0.081$ ,  $\hat{\rho} \approx 0.299$ – $0.301$ ). Regression adjustment gives the most reliable uncertainty quantification: large accepted set ( $n = 500$ ), tolerance-robust, and verifiable via sensitivity analysis. The SL posterior is the sharpest; synthetic truth recovery (Section 4.7) confirms the intervals are calibrated, though low ESS (7.5–11.9) makes the quoted standard deviations imprecise. ABC-MCMC at fixed  $\varepsilon$  achieves shrinkage comparable to rejection ABC, confirming that the tolerance rather than the sampler is the binding constraint; the  $\beta$ – $\rho$  correlation is preserved throughout. SMC-ABC resolves this by tightening  $\varepsilon$  iteratively, reaching 0.17/0.15/0.11 with the balanced set and 0.06/0.18/0.08 with the semi-automatic variant — a  $3\times$  improvement over ABC-MCMC on all parameters and a marked reduction in  $\beta$ – $\rho$  confounding (Table 10).

### 5.1 Simulator Budget

Table 9 breaks down simulator calls per method. Rejection ABC, semi-automatic ABC, LASSO ABC, and regression adjustment all share the same prior pool and incur no additional cost, so comparisons among them are like-for-like. Cross-family comparisons are not budget-neutral: the superior sharpness of SMC-ABC and synthetic likelihood partly reflects their substantially larger simulator investment.

### 5.2 Posterior Correlation

Table 10 reports the Pearson correlation  $\hat{\rho}(\beta, \rho)$  computed from each method’s posterior draws. Regression adjustment shows the weakest correlation (0.35), consistent with the linear correction partially disentangling the two parameters. SMC-ABC balanced and ABC-MCMC balanced are similar ( $\approx 0.75$ – $0.76$ ), confirming that iterative  $\varepsilon$  tightening reduces confounding but does not eliminate it; the semi-automatic variants push the correlation higher (0.84–0.94) because projecting onto three dimensions discards the rewiring and degree features that carry independent  $\rho$  signal.

### 5.3 Posterior Predictive Check

Figure 9 overlays 5th–95th percentile predictive bands on all three observed summaries for the SMC-ABC balanced posterior. The observed mean lies within the band throughout, confirming that the posterior is well-specified and not overconfident.



Table 8: Posterior standard deviations across all methods. Values in parentheses are shrinkage = posterior std / prior std. <sup>‡</sup>SL intervals are calibrated (Section 4.7) but low ESS (7.5–11.9) makes the quoted std imprecise.

| Method                                                                                     | $\sigma_\beta$ | $\sigma_\gamma$ | $\sigma_\rho$ | Notes                                |
|--------------------------------------------------------------------------------------------|----------------|-----------------|---------------|--------------------------------------|
| Prior                                                                                      | 0.130          | 0.052           | 0.231         | –                                    |
| <b>Rejection ABC</b>                                                                       |                |                 |               |                                      |
| Balanced ( $q = 1\%$ )                                                                     | 0.062          | 0.020           | 0.061         | $n = 100$                            |
| LASSO ( $q = 1\%$ )                                                                        | 0.062          | 0.020           | 0.061         | All 15 features kept                 |
| Semi-auto ( $q = 1\%$ )                                                                    | 0.023          | 0.017           | 0.051         | 3-d projection                       |
| <b>Regression adjustment</b>                                                               |                |                 |               |                                      |
| Balanced ( $q = 5\%$ )                                                                     | 0.024          | 0.007           | 0.008         | $n = 500$ , robust                   |
| <b>Synthetic likelihood<sup>‡</sup></b>                                                    |                |                 |               |                                      |
| Balanced                                                                                   | 0.003          | 0.002           | 0.005         | $N_{\text{sim}} = 150$ , accept. 24% |
| <b>ABC-MCMC (adaptive 0.5<math>\times</math>, <math>N = 30,000</math>)</b>                 |                |                 |               |                                      |
| Balanced                                                                                   | 0.0626 (0.48)  | 0.0204 (0.39)   | 0.0665 (0.29) | $\hat{R} < 1.001$                    |
| LASSO                                                                                      | 0.0700 (0.54)  | 0.0206 (0.40)   | 0.0691 (0.30) | $\hat{R} < 1.001$                    |
| Semi-auto                                                                                  | 0.0281 (0.22)  | 0.0376 (0.72)   | 0.0637 (0.28) | $\hat{R} < 1.001$                    |
| <b>SMC-ABC (refined 0.5<math>\times</math>, <math>T = 15</math>, <math>N = 100</math>)</b> |                |                 |               |                                      |
| Balanced                                                                                   | 0.0216 (0.17)  | 0.0078 (0.15)   | 0.0250 (0.11) |                                      |
| LASSO                                                                                      | 0.0238 (0.18)  | 0.0086 (0.16)   | 0.0286 (0.12) |                                      |
| Semi-auto                                                                                  | 0.0082 (0.06)  | 0.0091 (0.18)   | 0.0187 (0.08) |                                      |

Table 9: Simulator call budget. One “individual run” = one call to `simulate_fast()` ( $N = 200$  agents,  $T = 200$  steps). The prior pool (400 000 runs) is shared by all pool-based methods. <sup>‡</sup>SMC budget is a lower bound; actual cost depends on per-generation acceptance rates. <sup>‡</sup>Each SL step evaluates  $N_{\text{sim}} = 150$  datasets of  $R_{\text{sim}} = 40$  replicates, giving 6 000 simulator calls per MCMC step.

| Method                                                                                    | Additional runs | Cumulative total  |
|-------------------------------------------------------------------------------------------|-----------------|-------------------|
| Prior pool (shared by pool-based methods)                                                 | 400 000         | 400 000           |
| Rejection ABC / Semi-auto / LASSO / Reg. adj.                                             | 0               | 400 000           |
| ABC-MCMC (3 variants $\times$ 30 000 steps $\times$ 40 reps)                              | 3 600 000       | 4 000 000         |
| SMC-ABC (3 variants, $\geq 15$ gens $\times$ 100 particles $\times$ 40 reps) <sup>†</sup> | $\geq 540,000$  | $\geq 4,540,000$  |
| Synthetic likelihood (2 001 steps $\times$ 6 000 runs/step) <sup>‡</sup>                  | 12 006 000      | $\geq 16,546,000$ |

## 6 Conclusion

We have applied a progressive hierarchy of simulation-based inference methods to recover the three parameters ( $\beta, \gamma, \rho$ ) of an adaptive-network SIR model from  $R = 40$  replicate observations.

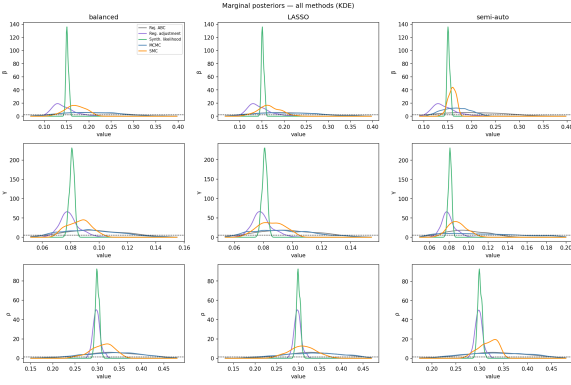
**Identifiability.** The clearest finding is that the three parameters are not equally identifiable from aggregate data.  $\gamma$  is the best-constrained: it directly controls epidemic duration and is strongly encoded in the infection time series, allowing even rejection ABC to shrink its posterior to 38% of the prior width.  $\beta$  and  $\rho$  are harder to separate because both modulate epidemic size through different mechanisms — higher transmission and lower rewiring produce similar infection curves — leaving a genuine positive posterior correlation between them that persists across all methods and summary sets. Rewiring summaries carry the independent signal needed to partially resolve this confounding; infection-curve summaries alone are essentially uninformative about  $\rho$ , and this structure is intrinsic to the observables rather than a sampler artefact.

**Method comparison.** Each method addresses a specific limitation of its predecessor. Semi-automatic ABC reduces summary dimensionality via ridge projection, tightening  $\beta$  and  $\rho$  at the cost of  $\gamma$  precision. Regression adjustment (Beaumont et al., 2002) dramatically sharpens all posteriors ( $\sigma_\rho$ : 0.087 $\rightarrow$ 0.008) and is

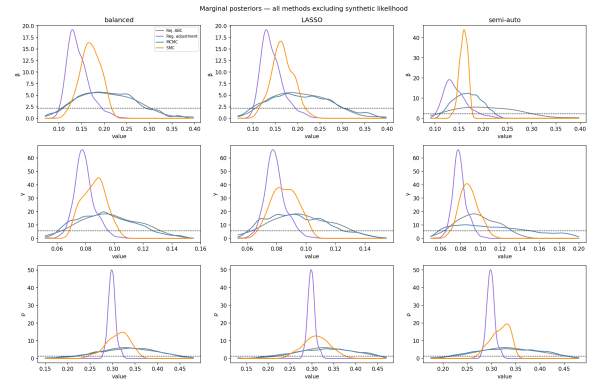


Table 10: Pearson correlation  $\hat{\rho}(\beta, \rho)$  for each posterior. Higher values indicate stronger  $\beta$ - $\rho$  confounding.

| Method                | $\hat{\rho}(\beta, \rho)$ | $n$    |
|-----------------------|---------------------------|--------|
| Rejection ABC         | 0.77                      | 100    |
| Regression adjustment | 0.35                      | 500    |
| Synthetic likelihood  | 0.78                      | 84     |
| ABC-MCMC balanced     | 0.75                      | 29 000 |
| ABC-MCMC LASSO        | 0.80                      | 29 000 |
| ABC-MCMC semi-auto    | 0.94                      | 29 000 |
| SMC balanced          | 0.76                      | 100    |
| SMC LASSO             | 0.78                      | 100    |
| SMC semi-auto         | 0.84                      | 100    |



(a) All methods including synthetic likelihood.



(b) All methods excluding synthetic likelihood.

Figure 8: Marginal posteriors (KDE) across all methods. Synthetic likelihood is shown separately as its extreme tightness compresses the scale for other methods.

robust to the choice of tolerance for  $q \lesssim 10\%$ . ABC-MCMC (Marjoram et al., 2003) achieves well-converged chains ( $\hat{R} < 1.001$ ) but its fixed  $\varepsilon$  leaves the  $\beta$ - $\rho$  confounding unresolved. SMC-ABC (Sisson et al., 2007) delivers the best overall performance, with the semi-automatic variant reaching shrinkage of 0.06 for  $\beta$  and 0.08 for  $\rho$ . Synthetic likelihood (Wood, 2010) yields the numerically sharpest posterior; synthetic truth recovery confirms the intervals are calibrated, though the low ESS (7.5–11.9) introduces uncertainty in the reported standard deviations.

**Limitations and future work.** SBC further confirms regression adjustment is well-calibrated across the full prior, not just at a single point. The low SL ESS could be raised by longer runs or a more targeted initialisation without materially changing the posterior location. For SMC-ABC, increasing the particle count beyond  $N = 100$  would reduce Monte Carlo noise in the final generation and give more stable shrinkage estimates. More broadly, the  $\beta$ - $\rho$  confounding is ultimately a limitation of the aggregate summary statistics: individual-level contact tracing data, if available, would provide direct information about rewiring events and likely resolve the degeneracy. Finally, neural simulation-based inference methods (e.g. Cranmer et al., 2020) that learn flexible approximations to the posterior directly from simulations represent a natural next step, bypassing the need for hand-crafted summary statistics altogether.

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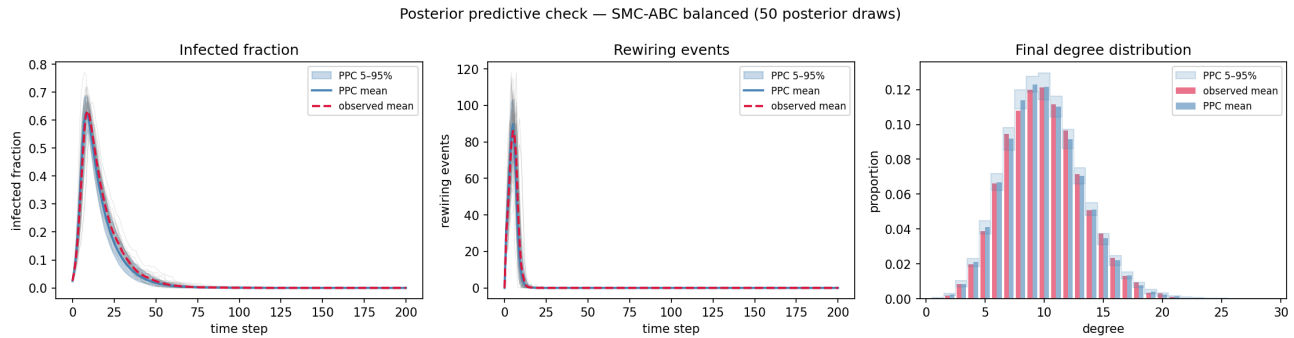


Figure 9: Posterior predictive check using SMC-ABC balanced (50 draws from the final generation, 40 replicates each). Shaded bands: 5th–95th percentile of per-draw replicate means. Blue line: predictive mean. Red dashed line: observed mean over 40 replicates. Grey traces: individual observed replicates. Left: infected fraction; centre: rewiring events; right: final degree distribution (normalised, bars = observed mean, line = predictive mean).

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